

MDARJ HR & Payroll System — Software Requirements Specification

MVP + Roadmap · Consolidated Edition v1.2 · Oct 3, 2026 · @trendow

Document information

This consolidated edition merges the original MVP SRS (Sep 30, 2026) with the Revised / Fixed Edition (Oct 1, 2026) and adds the release roadmap after the MVP. Where the two editions differ, the Oct 1 corrections apply.

Revision v1.2 (Oct 3, 2026) brings the SRS in line with the agreed decisions in `decisions.md` and the Nov 10, 2026 delivery plan. It replaces the Oct 1 consolidated text. The Sep 30 and Oct 1 PDF editions are superseded.

Source edition	Date	What it contributes
Revision v1.2	Oct 3, 2026	Decisions D-01 to D-03 applied; release split into MVP and MVP-2; missing settings named; open items linked to <code>assumptions.md</code> ; roadmap written out as text
Revised / Fixed Edition	Oct 1, 2026	Maximum 50 employees per company for the MVP; failed logins lock the user account only; user account status; Appendix B implementation baseline
Original MVP SRS	Sep 30, 2026	Full requirement wording and priorities, worked payroll example, Appendix A usage and severity, roadmap v1.1 to v4

The two corrections carried into every section:

- The MVP is designed, tested and optimized for companies with up to 50 employees.
- Failed login attempts lock the user account temporarily. They never delete, terminate, suspend or deactivate the employee record.

Changes in v1.2:

- Appendix B: the backend is NestJS and the frontend is Next.js (D-01), replacing Django + DRF and React + Vite.
- Weekly rotation shifts (FR-SH-3) move to v1.1 (D-02). Section 2.4 and the `shift_assignments` table are updated.
- The AI "Ask" chat (FR-AI-10) moves to v1.1 (D-03).
- New section 1.6 splits delivery into the MVP (Nov 10, 2026) and MVP-2 (Nov 16, 2026). Requirements that depend on MVP-2 say so.
- Settings that rules already used are now named: minimum overtime minutes (FR-CS-5), contract "expiring soon" days (FR-CT-2), AI daily limits (FR-AI-7, FR-AI-12).
- Section 2.5 links each open item to its working assumption in `assumptions.md`.
- Section 7.2 writes the roadmap out as text, with v1.1 updated.

1. Introduction

1.1 Purpose

This document specifies the first free version (MVP) of the MDARJ HR & Payroll System and the releases planned after it. It is the reference for design, development, testing and client sign-off. Anything not listed in sections 3 to 6 is out of scope for the MVP.

1.2 Product scope

The MVP covers the employee cycle from candidate to monthly salary in one web system:

- Company setup and policies
- Recruitment (candidates and interviews)
- Employee profiles and documents
- Contracts
- Shifts and roster
- Attendance and the attendance exceptions inbox
- Leave and public holidays
- Payroll (base salary + additions – deductions)
- AI compliance assistant powered by the OpenAI API (Egyptian Labor Law or company rules)

The MVP is offered free, so it favours simple, reliable features over advanced ones. Advanced features are planned in section 7. Recruitment and the AI compliance assistant are delivered in the second MVP release (MVP-2); see 1.6.

1.3 Definitions

Term	Meaning
Company	One client organisation (tenant). Its data is isolated from every other company.
Admin / HR Manager	The company user who configures the system, manages HR data, and approves requests.
Employee	A person employed by the company, with a self-service login.
Candidate	A person in the recruitment pipeline. Not an employee until converted.
Leave (إجازة)	Time off an employee requests (annual, sick, casual, unpaid, etc.).
Public holiday (عطلة رسمية)	A calendar day off for the whole company.
Shift template	A named working period (start, end, break, grace).
Roster	The generated schedule of which employee works which shift on which date.
Attendance record	One employee's check-in and check-out for one shift date.
Attendance exception	A deviation detected by the system (late, early leave, absence, overtime, missing check-out) that needs a manager decision.
Payroll run	The monthly calculation of all salaries for one company and one month.
Compliance status	Green (no open finding), yellow (at risk) or red (open violation finding).

1.4 References

- MDARJ HR & Payroll presentation, 28 July 2026 (scope marked by the client)
- Egyptian Labor Law No. 14 of 2025 (in force since 1 September 2025) and its executive regulations. The legal values in Appendix A are reference data only and must be confirmed against the official text and executive regulations with qualified legal review before launch.
- Egypt's Personal Data Protection Law No. 151 of 2020

1.5 Conventions

FR-XX-n = functional requirement, BR-n = business rule, NFR-n = non-functional requirement.

"Must" = required for the MVP. "Should" = required unless time runs short. "May" = optional.

Requirements marked "MVP-2" are delivered in the second MVP release (see 1.6). Requirements marked "v1.1" are moved out of the MVP and kept here for reference.

Currency is EGP. Dates use the Gregorian calendar.

1.6 Releases

Release	Target date	What it contains
MVP	Nov 10, 2026	Everything in sections 3 to 6 that is not marked MVP-2 or v1.1
MVP-2	Nov 16, 2026	Everything marked MVP-2: section 3.4 Recruitment, section 3.12 AI compliance, FR-CT-5, section 4.5 and NFR-9
Pilot	December 2026	One full month with a pilot company, from setup to payslips
v1.1	2–3 weeks after the MVP	FR-SH-3 and FR-AI-10, plus the v1.1 items in 7.2

The law reference in Appendix A (NFR-10) is loaded in the MVP, because FR-LV-2 pre-creates the statutory leave types from it.

Until MVP-2 ships, the screens that show recruitment or compliance data show an empty state: the employee Compliance tab (FR-EM-8), the dashboard compliance summary and candidate counts (FR-DB-1) and red-finding notifications (FR-DB-3).

2. Overall description

2.1 Product perspective

MDARJ HR is a multi-tenant web application (SaaS). Each company signs up, runs a setup wizard, and gets an isolated workspace. The system is used in a browser on desktop and phone; there is no native mobile app in the MVP.

2.2 User roles

There are no departments or reporting lines in the MVP, so every approval goes to an Admin / HR Manager. A company may have more than one Admin. For the MVP, Admin permissions are the same regardless of which Admin created the account.

Role	Who	Can do
Admin / HR Manager	Company owner or HR staff	Configure the company; manage employees, candidates, contracts, shifts, leave, payroll; approve all requests and exceptions; use the AI compliance assistant (MVP-2)
Employee	Every hired employee, including trainees	View own profile, check in and out, request leave, add reasons to own exceptions, view own payslips
Interviewer	A user named on an interview (MVP-2, with recruitment)	See the candidates assigned to them; record interview result, score and notes

2.3 Operating environment

- Latest two versions of Chrome, Edge, Safari and Firefox, on desktop and mobile.
- Arabic (right-to-left) and English interfaces, switchable per user.
- Cloud hosting with a managed PostgreSQL database and object/file storage for documents.
- Time zone set per company (default Africa/Cairo).

2.4 Constraints

- The MVP is free and is designed, tested and optimized for companies with up to 50 employees.
- No paid third-party services except the OpenAI API (a small, low-cost model) and basic hosting.
- No fingerprint device integration, native mobile app, GPS or QR check-in in the MVP.
- No notification emails or WhatsApp messages in the MVP. Only transactional emails are sent (invites and password reset).
- The system gives compliance guidance, not legal advice.
- Social insurance and income tax are entered as manual deduction lines; the system does not calculate them.
- Compliance warnings and suggestions come from the OpenAI API, working on facts the application calculates. Attendance labels, exceptions and payroll amounts remain deterministic application logic.
- Contracts show an in-app "expiring soon" badge; no email alert is required.
- Office IP restriction for check-in is included as an optional company setting.
- Only fixed shifts are supported in the MVP. Weekly rotations move to v1.1 (FR-SH-3, D-02).
- Employees join by an invite link sent by HR; there is no public sign-up for employees.

2.5 Assumptions to confirm before development

- Final legal values in Appendix A must be verified before production launch (A-01). FR-LV-2 already uses them in the MVP, so the review is needed before a real company uses the system.
- Decide whether the MVP stores an employee disability field for the statutory annual-leave rule (EG-05). Until decided, the field is not stored and the 45-day entitlement is not checked (D-04, A-02).
- Confirm the public-holiday overtime interpretation and the night-work definition used by the legal reference (EG-13). Until confirmed, public-holiday overtime uses the 2.00 multiplier and all overtime is classed as day overtime (A-03, A-04).
- The other working assumptions are listed in `assumptions.md`.

3. Functional requirements

3.1 Company setup and settings (CS)

ID	Requirement	Priority
FR-CS-1	A company owner signs up with company name, own name, email and password, and becomes the first Admin.	Must
FR-CS-2	After sign-up, a setup wizard walks the Admin through: company profile, work week, public holidays, job titles, shift templates, leave types, payroll settings, compliance mode, check-in settings. Each step can be skipped and finished later.	Must
FR-CS-3	Company profile holds name, logo, address, phone, time zone (default Africa/Cairo) and currency (EGP).	Must
FR-CS-4	Work week defines working days and weekly rest days (e.g. Friday and Saturday).	Must
FR-CS-5	Payroll settings define: daily rate formula (monthly salary ÷ 30 by default, or ÷ 26, or ÷ working days of the month); hourly rate (BR-6); lateness policy (exact per minute, or tiered bands); overtime multipliers; minimum overtime minutes (default 30, BR-5).	Must
FR-CS-6	Compliance mode is a yes/no choice: "Follow the Egyptian Labor Law". The Admin can also write company rules as free text (see 3.12).	Must
FR-CS-7	Check-in settings: allowed check-in window before shift start (default 60 min) and an optional list of allowed office IP addresses.	Should
FR-CS-8	Every setting can be changed later from Settings. A change applies from its effective date and never alters an approved payroll run.	Must
FR-CS-9	Job titles are a company list managed in Settings and picked on each employee.	Must

3.2 Users and access (UA)

ID	Requirement	Priority
FR-UA-1	Employees and interviewers cannot sign up on their own. The Admin creates them and the system issues an invite link, valid 7 days, to set a password.	Must
FR-UA-2	The invite link is sent by email when the person has one, and can always be copied so HR can share it by WhatsApp or SMS.	Must
FR-UA-3	Users log in with email or phone number plus password. Workers without email use their phone number.	Must
FR-UA-4	Password reset: by email link when an email exists; otherwise the Admin generates a new invite link.	Must
FR-UA-5	Permissions are enforced on the server. An Employee sees only their own data; an Interviewer sees only the candidates on their interviews.	Must
FR-UA-6	Terminating an employee deactivates their login on the termination date.	Must
FR-UA-7	A company can have several Admins.	Should
FR-UA-8	User account status (Invited, Active, Locked, Deactivated) is distinct from employee employment status. Failed-login locking changes user access only; it never deletes, terminates, suspends or deactivates the employee record.	Must

3.3 Employees (EM)

ID	Requirement	Priority
FR-EM-1	The Admin creates an employee manually or, from MVP-2, by converting a hired candidate (3.4). The system assigns an employee code.	Must
FR-EM-2	Personal data: full name (Arabic and English), national ID number, date of birth, gender, phone, email, address, emergency contact name and phone, profile photo.	Must
FR-EM-3	Job data: job title, employment type (full-time, part-time, trainee), hire date, probation end date, training end date (trainees), assigned shift, base salary with effective date.	Must
FR-EM-4	Salary history: every base salary change is kept with its effective date; payroll uses the salary valid in each period.	Must
FR-EM-5	Recurring pay items: fixed monthly allowances or deductions per employee (e.g. transport allowance) that are added to every payroll run automatically.	Should
FR-EM-6	Documents: upload PDF, JPG or PNG files (max 10 MB each) with a document type and an optional expiry date.	Must
FR-EM-7	Employee status: Active, Suspended or Terminated. Termination records date and reason.	Must
FR-EM-8	The employee profile shows tabs: Personal, Job, Contract, Documents, Attendance, Leave, Payroll, Compliance. Until MVP-2, the Compliance tab shows an empty state.	Must
FR-EM-9	The employee list can be searched and filtered by status, employment type, job title and shift, and exported to Excel.	Must
FR-EM-10	Employees can view their own profile and request corrections; only the Admin edits data.	Should

3.4 Recruitment (RC)

Release: MVP-2. There are no job postings in the MVP. Candidates are added by the Admin and tracked through statuses.

ID	Requirement	Priority
FR-RC-1	Candidate data: full name, phone, email, position applied for, source (referral, walk-in, social media, other), CV file, expected salary, date added.	Must
FR-RC-2	Candidate statuses: New, Screening, Interview scheduled, Interviewed, Offer, Hired, On hold, Rejected, Withdrawn.	Must
FR-RC-3	Candidates are shown on a board with one column per status (drag to move) and in a searchable list.	Must
FR-RC-4	Every status change is logged with date, user and an optional comment. Rejected requires a reason; On hold takes an optional follow-up date.	Must
FR-RC-5	Notes: any number of dated notes per candidate, each showing its author.	Must
FR-RC-6	Interviews: several per candidate, each with date and time, type (in person, phone, online), interviewer(s), place or link, result (Pending, Passed, Failed), score 1-5 and notes.	Must

ID	Requirement	Priority
FR-RC-7	The system warns when a new candidate has the same phone or email as an existing one.	Should
FR-RC-8	Rejected and On hold candidates stay searchable and can be moved back into the pipeline.	Must
FR-RC-9	"Convert to employee" on a Hired candidate creates an employee pre-filled with the candidate's data and keeps the link to the candidate record.	Must

3.5 Contracts (CT)

ID	Requirement	Priority
FR-CT-1	The Admin uploads a contract file (PDF, JPG or PNG, max 10 MB) to an employee and enters type (indefinite, fixed-term, trainee, part-time), start date, end date (required for fixed-term), agreed salary and notes.	Must
FR-CT-2	The system computes contract status: Upcoming, Active, Expiring soon (end date within the company's "expiring soon" days; company setting, default 30), Expired.	Must
FR-CT-3	An employee has at most one Active contract at a time. A renewal creates a new contract linked to the previous one; history is kept.	Must
FR-CT-4	The dashboard shows counts of Expiring soon and Expired contracts, and the contracts list can be filtered by status.	Must
FR-CT-5	An expired contract of an Active employee creates a red compliance finding (3.12).	Must · MVP-2

3.6 Shifts and roster (SH)

The Admin sets shifts up once; the system then builds the daily roster on its own. The Admin only handles exceptions.

ID	Requirement	Priority
FR-SH-1	Shift template fields: name, start time, end time, crosses midnight (yes/no), unpaid break minutes, late grace minutes, early-leave grace minutes, working days of the week.	Must
FR-SH-2	Fixed assignment: an employee is assigned one shift template from a start date.	Must
FR-SH-3	Weekly rotation: an employee or group follows an ordered list of templates that repeats every N weeks from a start date (e.g. week 1 morning, week 2 evening, week 3 night). Moved to v1.1 (D-02); not built in the MVP.	v1.1
FR-SH-4	A nightly job generates roster entries (employee, shift date, shift template, expected start and end) for the next 14 days, and regenerates them when an assignment changes.	Must
FR-SH-5	Roster generation marks weekly rest days, public holidays and approved leave instead of a working shift.	Must
FR-SH-6	The Admin can override one roster day (another shift, or a day off) with a reason; the change is logged.	Must

ID	Requirement	Priority
FR-SH-7	A weekly roster grid shows employees by days; employees see their own upcoming shifts.	Must

3.7 Attendance (AT)

ID	Requirement	Priority
FR-AT-1	The employee portal shows a Check in / Check out button for today's shift. The server time is recorded, never the device time; the IP address is stored with each punch.	Must
FR-AT-2	Check-in is accepted from the configured window before shift start until shift end. If an IP allow-list is set, a punch from another IP is refused with a clear message.	Must
FR-AT-3	A check-out closes the open record for that shift. A night shift record belongs to its shift date, not the calendar date of the check-out.	Must
FR-AT-4	After each shift ends (plus 2 hours), the system labels the day: Present, Late, Early leave, Absent, Overtime, Missing check-out, On leave, Rest day, Public holiday. A day can carry more than one label (e.g. Late and Early leave).	Must
FR-AT-5	The Admin can add or correct a punch with a mandatory reason; the original values and the change are kept in the audit log.	Must
FR-AT-6	Views: a live "today" board (in, late, not yet in, absent, on leave); a monthly calendar per employee with totals of late minutes, absent days and overtime hours.	Must
FR-AT-7	Monthly attendance can be exported to Excel.	Should

3.8 Attendance exceptions inbox (EX)

The system detects, the manager decides, the system calculates. Approved exceptions become payroll lines automatically.

ID	Requirement	Priority
FR-EX-1	The system creates an exception for each Late, Early leave, Absent, Overtime and Missing check-out label that is beyond the grace minutes.	Must
FR-EX-2	Each exception shows employee, date, type, minutes or days, the suggested amount in EGP (negative for deductions, positive for overtime) and the policy used to compute it.	Must
FR-EX-3	The employee can add a reason and an attachment to their own exception within 3 days of its creation.	Must
FR-EX-4	The Admin chooses: Approve, Reject, Edit amount and approve, or Convert to leave (pick a leave type; the time is taken from that balance instead of the salary). A note is optional.	Must
FR-EX-5	For Missing check-out, the Admin enters the actual check-out time and the system recalculates the day.	Must
FR-EX-6	Bulk approve and bulk reject work on a filtered selection (e.g. all late arrivals under 30 minutes this month).	Must
FR-EX-7	Each decision stores who decided, when, and the final amount.	Must

ID	Requirement	Priority
FR-EX-8	An exception decided after its month's payroll is approved goes to the next month's run as a carried-forward line.	Must

3.9 Leave (LV)

Each company defines its own leave types and request rules; the system enforces them before the Admin sees a request.

ID	Requirement	Priority
FR-LV-1	Leave type fields: name (Arabic and English), paid (yes/no), yearly entitlement in days, counted as working days or calendar days, minimum notice in hours (e.g. 48 or 72), exempt from notice (yes/no, for sick and emergency leave), maximum consecutive days, eligible after N months of service, attachment required (yes/no), gender restriction, maximum times per employment, carry-over allowed and its maximum days.	Must
FR-LV-2	When law mode is on, the system pre-creates the statutory leave types from Appendix A; the Admin may make them more generous but not less.	Must
FR-LV-3	Balances are kept per employee, per leave type, per year. Opening balances can be entered when a company starts using the system.	Must
FR-LV-4	The yearly entitlement is granted on 1 January or on the eligibility date, optionally pro-rated by the months left in the year. Unused days carry over up to the type's maximum.	Must
FR-LV-5	An employee requests leave with type, start date, end date, full or half day, reason and attachment.	Must
FR-LV-6	The system validates the request against the rules in BR-10 to BR-15 and refuses it with the reason when a rule fails.	Must
FR-LV-7	The Admin approves or rejects with an optional note. On approval the balance is deducted and the roster days are marked On leave.	Must
FR-LV-8	The employee can cancel a pending request. The Admin can cancel an approved future leave; the balance is restored.	Must
FR-LV-9	A leave calendar shows who is off on which days.	Should

3.10 Public holidays (PH)

ID	Requirement	Priority
FR-PH-1	Each company keeps a public holidays calendar per year: name, date or date range, paid (yes/no).	Must
FR-PH-2	The system offers the Egyptian official holidays as a starting list; religious holidays whose dates vary are marked "date to confirm" until the Admin confirms them.	Must
FR-PH-3	Public holidays are excluded from working-day leave counts and from expected attendance.	Must
FR-PH-4	Work on a public holiday creates an Overtime exception at the holiday multiplier.	Must

3.11 Payroll (PR)

The MVP formula is: net salary = base salary + additions – deductions. Social insurance and income tax are manual deduction lines. All payroll amounts are calculated by deterministic backend business logic. AI may review facts and rules but never calculates, approves, changes or writes payroll amounts.

ID	Requirement	Priority
FR-PR-1	The Admin creates one payroll run per month. It includes every employee who was Active on any day of that month.	Must
FR-PR-2	Base salary is pro-rated by days for employees who joined or left during the month, using the company's daily rate formula.	Must
FR-PR-3	Additions are pulled automatically from approved exceptions (overtime) and recurring allowances; deductions from approved exceptions (lateness, early leave, absence) and unpaid leave days.	Must
FR-PR-4	The Admin can add manual lines per employee: bonus, allowance, penalty, advance repayment, social insurance, income tax, other; each with a label and an amount.	Must
FR-PR-5	Before approval the run shows warnings: pending exceptions for the month, employees with no base salary, negative net salaries.	Must
FR-PR-6	Run states: Draft, Approved (locked), Paid (optional payment date). Only an Admin can reopen an approved run, with a mandatory reason that is logged.	Must
FR-PR-7	Each employee gets a payslip (company, employee, month, base, every addition and deduction line, net) as a PDF. Employees see their own payslips once the run is approved.	Must
FR-PR-8	The payroll register lists every employee's base, additions, deductions and net with column totals, exportable to Excel and PDF.	Must

3.12 AI compliance and assistant (AI)

The system calculates the facts; the OpenAI API reads them with the law or company rules and returns warnings and suggestions. Attendance labels, exceptions and payroll amounts are always calculated in code, never by the AI. The AI never changes data.

Release: MVP-2, except FR-AI-10, which moves to v1.1.

ID	Requirement	Priority
FR-AI-1	Integration with the OpenAI API using a small, low-cost model (for example GPT-5 mini) with structured JSON output. The model name is a setting, so it can be changed without a release.	Must
FR-AI-2	Every night the system builds a fact sheet per active employee from existing data: employment type, months of service, age band, contract type and end date, probation end, scheduled hours per day and week, rest days in the last 7 days, break minutes, leave granted and taken by type, overtime hours and the multiplier paid, maternity leave dates.	Must

ID	Requirement	Priority
FR-AI-3	Rules given to the AI: with law mode on, the versioned law reference in Appendix A plus the company's rules text; with law mode off, the company's rules text only. The Admin writes company rules as free text and they are sent as written.	Must
FR-AI-4	Nightly check: fact sheets are sent in batches (e.g. 20 employees per call) with the rules. The AI returns findings: employee code, rule, severity (yellow or red), message, suggested action, and the facts it relied on.	Must
FR-AI-5	Guardrail: the system checks that every fact a finding quotes matches the fact sheet. A finding that quotes a wrong value is discarded and logged.	Must
FR-AI-6	Policy check: when the Admin saves leave types, shift templates or payroll settings, the AI compares them with the rules and returns warnings at once (e.g. annual leave set below the legal minimum).	Must
FR-AI-7	An employee's status is the worst open finding: green with none, yellow at risk, red on a violation. The company status summarises the counts. The Admin can press "Run check now" a limited number of times a day (system setting, proposed default 3; A-32).	Must
FR-AI-8	Finding states: Open, Resolved (the next run no longer reports it), Dismissed (with a reason). A new red finding creates an in-app notification. The dashboard and the employee Compliance tab list findings, each marked "AI suggestion, review before acting".	Must
FR-AI-9	An "Explain" button on a finding returns a plain-language explanation and next step in the user's language.	Must
FR-AI-10	The Admin can ask questions in Arabic or English. The assistant answers from the company rules, the law reference and the fact sheets, names the rule it relied on, and is read-only. Moved to v1.1 (D-03); not built in the MVP.	v1.1
FR-AI-11	Privacy: the AI receives employee codes, not names. National ID numbers, contact details, salary amounts and uploaded files are never sent. The system adds names back when it shows results.	Must
FR-AI-12	Every AI output shows that it is guidance, not legal advice. AI calls per company per day are capped (system setting, proposed default 200; A-32).	Must

3.13 Dashboard, notifications and audit (DB)

ID	Requirement	Priority
FR-DB-1	Admin dashboard: today's attendance counts, pending leave requests, pending exceptions, contracts expiring or expired, compliance summary, candidates by status. The compliance summary and candidates by status are filled from MVP-2.	Must
FR-DB-2	Employee home: today's shift and check-in button, leave balances, status of own requests, latest payslip.	Must
FR-DB-3	In-app notification centre with unread count, for: leave requested or decided, exception created or decided, payslip available, red finding (from MVP-2), contract expiring.	Must
FR-DB-4	An audit log records who changed what and when for salaries, punches, approvals, payroll runs and settings.	Must

4. Business rules

Every amount the system suggests follows from the rules below and the company's settings. Values marked "company setting" are examples, not law.

4.1 Attendance labels

ID	Rule
BR-1	Late: check-in after shift start + late grace. Late minutes are counted from shift start.
BR-2	Early leave: check-out before shift end – early-leave grace. Early minutes = shift end – check-out.
BR-3	Absent: no check-in by shift end on a working roster day without approved leave.
BR-4	Missing check-out: a check-in with no check-out 2 hours after shift end.
BR-5	Overtime: time worked after shift end of at least the minimum overtime minutes (company setting, default 30), and any time worked on a rest day or public holiday. Arriving early is not overtime.

4.2 Calculations

ID	Rule
BR-6	Daily rate = monthly base salary ÷ divisor (30, 26 or working days of the month; company setting). Hourly rate = daily rate ÷ (shift hours – unpaid break).
BR-7	Lateness and early-leave deduction. Exact mode: minutes × hourly rate ÷ 60. Tiered mode: company bands, for example 16–60 min = ¼ day, 61–120 min = ½ day, over 120 min = 1 day.
BR-8	Absence deduction suggestion = 1 daily rate per absent day. A multiplier above 1 counts as a penalty and raises a yellow finding for review.
BR-9	Overtime amount = overtime hours × hourly rate × multiplier. With law mode on, multipliers come from Appendix A.

Worked example: shift 11:00–19:00 (no break), base salary 9,000 EGP, divisor 30. Daily rate = 300 EGP; hourly rate = 37.50 EGP. The employee checks in at 11:00 and out at 14:00, so early leave = 5 h and the suggested deduction = $5 \times 37.50 = 187.50$ EGP.

4.3 Leave validation

ID	Rule
BR-10	Notice: leave start – request time ≥ minimum notice hours, unless the leave type is exempt from notice.
BR-11	Balance: requested days ≤ available balance for paid types with an entitlement. Unpaid leave has no balance check but respects its maximum days.
BR-12	No overlap with another pending or approved leave of the same employee.
BR-13	Eligibility: months of service ≥ the type's minimum; gender restriction; maximum times per employment.
BR-14	Requested days ≤ the type's maximum consecutive days.

ID	Rule
BR-15	Day counting: working-day types skip rest days and public holidays; calendar-day types count every day. A required attachment must be present.

4.4 Payroll

ID	Rule
BR-16	Only approved exceptions enter a run. If the Admin approves a run with pending exceptions, those carry forward to the next month.
BR-17	An approved run is locked. Later corrections for that month reach the next run as carried-forward lines, never by editing the locked run.
BR-18	Net = pro-rated base + additions – deductions, rounded to 2 decimals. A negative net blocks approval until the Admin confirms it.

4.5 Compliance

Release: MVP-2.

ID	Rule
BR-19	Red = an open finding that a rule is broken now. Yellow = an open finding that a rule is at risk soon (e.g. a contract ending within 30 days, weekly hours within 4 h of the maximum). The thresholds are written in the versioned rules given to the AI.
BR-20	A finding stays open until a later run no longer reports it, or the Admin dismisses it with a reason. The same finding (employee + rule) is never duplicated between runs.
BR-21	AI findings never change data or amounts. Any correction (e.g. adding leave days, paying an overtime difference) is made by the Admin in the relevant module.

5. Data model

The MVP uses approximately 30 relational tables. Every company-owned table carries `company_id`, and every server-side read and write is scoped to the authenticated user's company, so one company can never read another's data. `law_references` is shared, versioned reference data.

5.1 Company, users and employees

Table	Key fields
companies	name, logo, time_zone, currency, law_mode, company_rules_text, settings (daily-rate divisor, lateness policy, overtime minimum, contract expiring-soon days, check-in window, IP allow-list)
users	email, phone, password_hash, role (admin, employee, interviewer), employee_id, status (invited, active, locked, deactivated), invite_token, invite_expires_at, language, failed_login_attempts, locked_until
job_titles	company_id, name_ar, name_en

Table	Key fields
employees	code, name_ar, name_en, national_id, birth_date, gender, phone, email, address, emergency_name, emergency_phone, photo, job_title_id, employment_type, hire_date, probation_end, training_end, status, termination_date, termination_reason, candidate_id
salary_history	employee_id, base_salary, effective_from
recurring_pay_items	employee_id, kind (addition, deduction), label, amount, valid_from, valid_to
documents	owner_type (employee, candidate), owner_id, doc_type, file_key, expiry_date, uploaded_by
contracts	employee_id, type, start_date, end_date, agreed_salary, file_key, previous_contract_id, notes

5.2 Recruitment

Table	Key fields
candidates	name, phone, email, position_applied, source, cv_file_key, expected_salary, status, on_hold_until, rejection_reason, employee_id
candidate_status_log	candidate_id, from_status, to_status, changed_by, changed_at, comment
candidate_notes	candidate_id, author_id, body, created_at
interviews	candidate_id, scheduled_at, type, place_or_link, result, score, notes
interview_interviewers	interview_id, user_id

When a hired candidate is converted, candidates.employee_id and employees.candidate_id point to each other, so both histories stay linked.

5.3 Shifts, attendance and leave

Table	Key fields
shift_templates	name, start_time, end_time, crosses_midnight, break_min, late_grace_min, early_grace_min, working_days
shift_assignments	employee_id, kind (fixed; rotation from v1.1), template_ids (ordered), cycle_weeks (v1.1), start_date, end_date
roster_entries	employee_id, shift_date, template_id, expected_start, expected_end, day_type (work, rest, holiday, leave), override_reason
attendance_records	employee_id, roster_entry_id, shift_date, check_in_at, check_out_at, check_in_ip, check_out_ip, labels, late_min, early_min, overtime_min, source (self, admin)
attendance_exceptions	attendance_record_id, employee_id, type, minutes, days, suggested_amount, final_amount, policy_snapshot, employee_reason, attachment_key, status, decided_by, decided_at, note, leave_request_id, payroll_run_id
leave_types	name_ar, name_en, paid, entitlement_days, count_mode, min_notice_hours, notice_exempt, max_consecutive, eligible_after_months, attachment_required, gender, max_times, carry_over_max, statutory

Table	Key fields
leave_balances	employee_id, leave_type_id, year, opening, granted, used, carried_over
leave_requests	employee_id, leave_type_id, start_date, end_date, half_day, days, reason, attachment_key, status, decided_by, decided_at, note
public_holidays	name, start_date, end_date, paid, confirmed

5.4 Payroll

Table	Key fields
payroll_runs	month, status (draft, approved, paid), approved_by, approved_at, paid_at
payroll_items	run_id, employee_id, base_prorated, total_additions, total_deductions, net
payroll_lines	payroll_item_id, kind (addition, deduction), category, label, amount, source_type (exception, recurring, manual, leave, carry-forward), source_id

5.5 Compliance, AI and system

Table	Key fields
law_references	code (e.g. EG-LABOR), version, effective_from, text (the rule list given to the AI); shared by all companies
compliance_runs	run_type (nightly, manual, policy), model, started_at, finished_at, tokens_in, tokens_out, status
compliance_findings	run_id, employee_id, rule_ref, severity, message, suggestion, evidence, state, first_seen, resolved_at, dismissed_reason
ai_usage	usage_date, calls, tokens_in, tokens_out
notifications	user_id, type, payload, read_at
audit_log	user_id, entity, entity_id, action, before, after, created_at

6. Non-functional requirements

ID	Area	Requirement
NFR-1	Tenant isolation	Every read and write is scoped to the user's company on the server. Automated tests prove one company cannot reach another's records or files.
NFR-2	Security	Passwords hashed with bcrypt or Argon2; HTTPS only; sessions expire after 12 hours of inactivity; 5 consecutive failed logins lock the user account for 15 minutes. Failed login attempts must never delete, terminate, suspend or deactivate the employee record.
NFR-3	Privacy	National ID, salary and documents are visible only to Admins and the employee concerned. Files are served through short-lived signed links. Personal data is handled in line with Egypt's Personal Data Protection Law No. 151 of 2020.

ID	Area	Requirement
NFR-4	Performance	Pages load in under 2 seconds for a company of up to 50 employees; a payroll run for 50 employees completes in under 10 seconds under normal MVP conditions.
NFR-5	Reliability	Daily database backups kept for 30 days. Nightly jobs (roster, attendance labels, compliance) are safe to re-run without creating duplicates.
NFR-6	Localisation	Full Arabic (right-to-left) and English interfaces; dates, numbers and EGP amounts formatted per language.
NFR-7	Usability	Works on phone screens from 360 px wide; check-in takes no more than 2 taps after login.
NFR-8	Auditability	Changes to salaries, punches, approvals, payroll and settings are written to an immutable audit log that users cannot edit or delete.
NFR-9	AI cost control	AI calls per company per day are capped, and checks run once a night in batches with a small model. If OpenAI is unavailable, the last results stay on screen with their date and every other module keeps working. Applies from MVP-2.
NFR-10	Maintainability	The law reference given to the AI is stored as versioned data, so a legal change is a data update, not a code release.

7. Out of scope and roadmap

7.1 Not in the MVP

- Departments, sections, reporting lines and org chart
- Job postings and online application forms
- Fingerprint or barcode device integration
- Native mobile app, GPS or QR check-in
- Social insurance and income tax calculation
- Multiple salary structures
- Accounting integration, journal entries and cost centres
- Analytics reports and payroll distribution by department
- Email and WhatsApp notifications (except invites and password reset)
- Multi-level approvals
- Weekly rotation shifts and the AI "Ask" chat (moved to v1.1)

7.2 Roadmap

Four releases follow the MVP, moving from quick wins (v1.1, 2–3 weeks after the MVP) to new markets and paid tiers (v4).

MVP (free): this document, delivered as the MVP and MVP-2 (see 1.6).

v1.1 Quick wins: 2–3 weeks after the MVP

- Weekly rotation shifts (FR-SH-3, moved from the MVP)
- AI "Ask" chat assistant (FR-AI-10, moved from the MVP)

- Excel import of fingerprint logs
- Bulk employee import from Excel
- Email notifications

v2 Operations: mobile and payroll depth

- Android app: GPS and QR check-in
- Kiosk check-in tablet
- Social insurance and income tax
- Loans and salary advances
- Departments and org chart
- Multi-level approvals
- Shift swap requests
- WhatsApp notifications

v3 Intelligence: AI and integrations

- Analytics dashboard
- AI contract reader
- AI CV screening
- Performance reviews
- Accounting integration
- Cost centres
- Multiple salary structures
- Public careers page

v4 Expansion: new markets, paid tiers

- Saudi, UAE and Jordan law packs
- End-of-service workflow
- Paid plans
- E-signature for contracts
- Training management

Each release builds on the MVP's data model; the law reference design lets v4 add new countries as data, not code.

Appendix A. Egyptian Labor Law reference for the AI (EG-LABOR v1)

The AI compliance check is given the rules below as its law reference, built from Labor Law No. 14 of 2025, in force since 1 September 2025. The values come from a published summary and are reference data only. Confirm each against the official text and the executive regulations with a labor lawyer before launch.

Code	Rule	Value	How the system uses it	Severity
EG-01	Daily hours	Max 8 hours, excluding breaks	Shift template or roster day longer than 8 working hours	Red
EG-02	Weekly hours	Max 48 hours	Scheduled week over 48 h; within 4 h of the limit	Red / Yellow
EG-03	Rest breaks	At least 1 hour a day; no more than 5 continuous working hours	Shift over 5 h with a break under 60 min	Red
EG-04	Weekly rest	At least 24 consecutive hours, fully paid	Roster with no rest day in 7 days	Red
EG-05	Annual leave	15 days in year 1 (after 6 months); 21 days from year 2; 30 days after 10 years of service or age 50; 45 days for employees with disabilities	Granted balance below the entitlement for the employee's service and age	Red
EG-06	Casual leave	6 days a year, max 2 consecutive	Pre-created leave type; company setting below this	Red
EG-07	Sick leave	3 months at 100%, 6 months at 85%, 3 months at 75%; certified; paid through social insurance	Pre-created leave type; information for the assistant	—
EG-08	Maternity leave	4 months paid, no minimum service, up to 3 times	Pre-created leave type; a refused or shortened request	Red
EG-09	After childbirth	No overtime during pregnancy or for 6 months after birth; 1 hour less a day from the 6th month of pregnancy	Overtime approved within 6 months after a maternity leave	Red
EG-10	Paternity leave	1 paid day on the day of birth, up to 3 times	Pre-created leave type	—
EG-11	Nursing breaks	2 breaks of 30+ minutes a day for 2 years after childbirth	Information for the assistant	—
EG-12	Hajj leave	1 month, full pay, once, after 5 years of continuous service	Pre-created leave type	—
EG-13	Overtime pay	Day +35%; night +70%; weekly rest day double pay plus a compensatory rest day; public holiday: day's wage plus double pay, or a compensatory day off	Default multipliers 1.35, 1.70, 2.00; an approved overtime amount below the multiplier	Red
EG-14	Probation	Max 3 months, once with the same employer	Probation end more than 3 months after hire date	Red
EG-15	Notice period	3 months for indefinite contracts	Information for the assistant (end-of-service is v4)	—
EG-16	Public holidays	13 official paid days a year; Islamic dates follow moon sighting	Preloads the public holidays calendar	—

Two checks are system rules, not law: an Active employee with an expired contract is red, and a contract ending within 30 days is yellow.

Open items:

- EG-05 needs a "disability" field on the employee to apply the 45-day entitlement. Decide whether to store it in the MVP. Still open (D-04); until decided, the 45-day entitlement is not checked (A-02).
- EG-13: confirm how the public holiday rate is applied (total 2× or 3× the hourly wage) and which hours count as night work. Working assumptions until confirmed: A-03 and A-04.
- EG-09: the MVP does not store pregnancy dates, so only the post-birth overtime rule is checked.

Sources: ZenHR, Egypt Labor Law at a Glance (EG-01 to EG-16) · Clyde & Co, New Labour Law issued in Egypt (EG-09).

Appendix B. Developer implementation baseline

This section translates the SRS into implementation constraints. It does not add product scope. The stack follows decision D-01 (Oct 1, 2026). Implementation choices still marked Proposed, such as the ORM and the job library, are in `decisions.md`.

Area	Baseline
Backend	NestJS (TypeScript)
Frontend	Next.js (TypeScript, App Router) + Tailwind CSS
Database	PostgreSQL
Background jobs	Redis-backed job queue with scheduled, retryable jobs (BullMQ proposed, D-23)
File storage	S3-compatible object storage; the database stores file metadata/keys
AI	OpenAI API behind a dedicated compliance service with structured output validation and rate/cost limits
Multi-tenancy	company_id on every company-owned table; server-side tenant scoping is mandatory

Implementation rules:

- Core business calculations must be deterministic backend/domain logic; AI is advisory and read-only.
- Use effective-dated settings and salary history so approved payroll is reproducible.
- All scheduled jobs must be idempotent and safe to rerun.
- Testing must include tenant isolation, permissions, payroll calculations, attendance edge cases, leave validation and locked payroll behaviour.
- Build Arabic/English and RTL into the frontend foundation rather than adding localisation after all screens are finished.